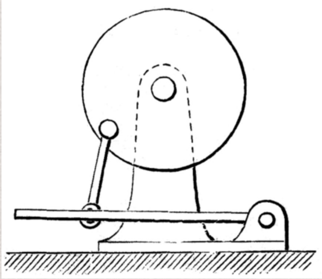
RESEARCH & INITIAL SKETCHES

Research



Advantage: Known for its quick-return, it really efficiently turns rotational motion into lateral motion using gravity to increase the speed down the wheel and decreases idle time.

Predicted Failure Modes:

1. Connections between wheel → arm and arm → pedal
2. Compactness → limited to 3 planes for 3 connections
3. Jamming in joints

Application:

- Sewing machines
    - Treadle flywheel system
- Shaping machines
- Textile and packing machinery
- Slotting machines

Driver Questions

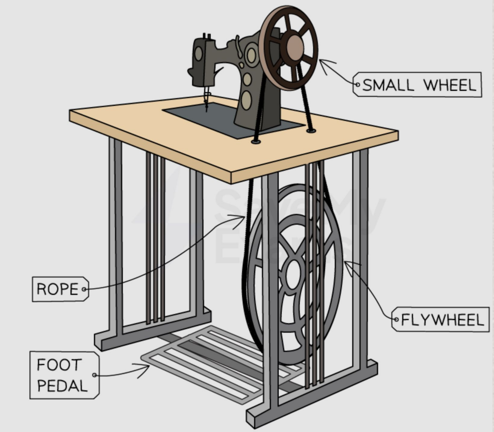
How would we maximize the ease of usability? How do we maximize the efficiency of transferring rotational motion into linear motion? How should we maximize "smoothness"? **How does the speed of pushing the pedal affect the mechanism?**

Metrics of Success

Ease of usability: The "smoothness" from pushing on the pedal
Durability: Works regardless of amount of force applied
Variety of Speeds: The mechanism works regardless of the speed of pressing down on the pedal

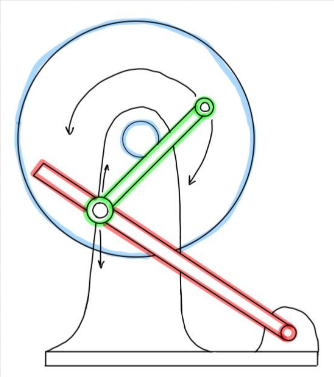
Our Use-Case: Pedaled treadle flywheel system

- Huge historical impact
- Standard mechanism for domestic sewing machines from mid-1800s to the early 1900s.
- Millions of machines were sold → machine *had* to be easy to use



CHOSEN USE CASE & PROTOTYPES

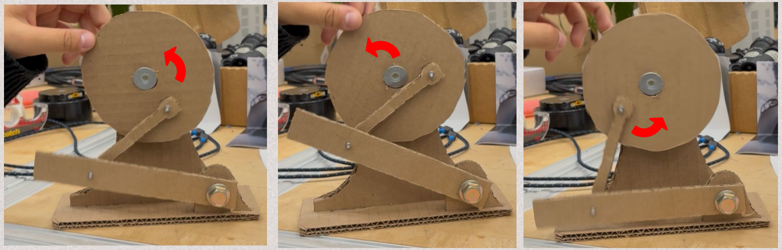
Sketching & Planning



How we determined our mechanism worked

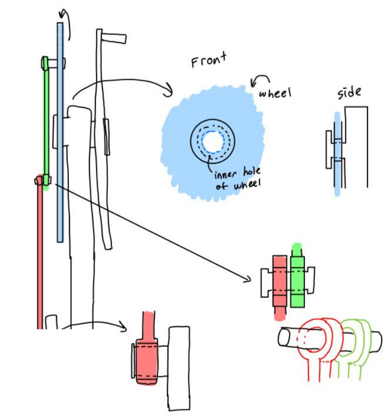
1. Push red lever arm up until green passed the center
2. Gravity had the arm fall & reset

Prototype 1

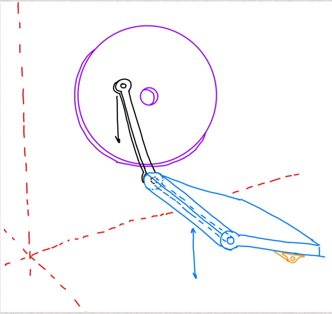


A very quick cardboard + scrap screws prototype. We found a couple issues & places we needed to focus on → extruding joints, weight distribution, joint connections

Prototype Joints
The right image shows the various joint ideas that initial arose during brainstorming. A combination of bushings and rods, similar to what we have now.

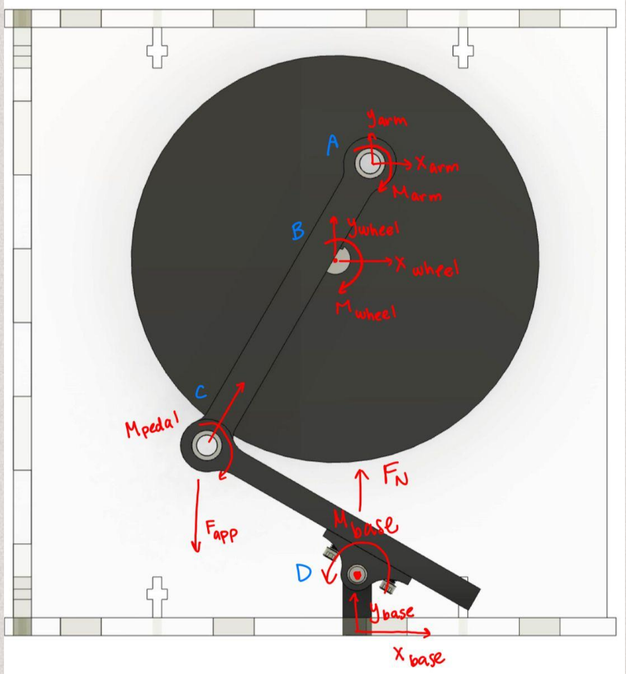


Prototype 2 – Mechanism



Focused mechanism prototype—we figured out all joints depended on the wheel to work so we focused on that.

FBD & Build Assembly

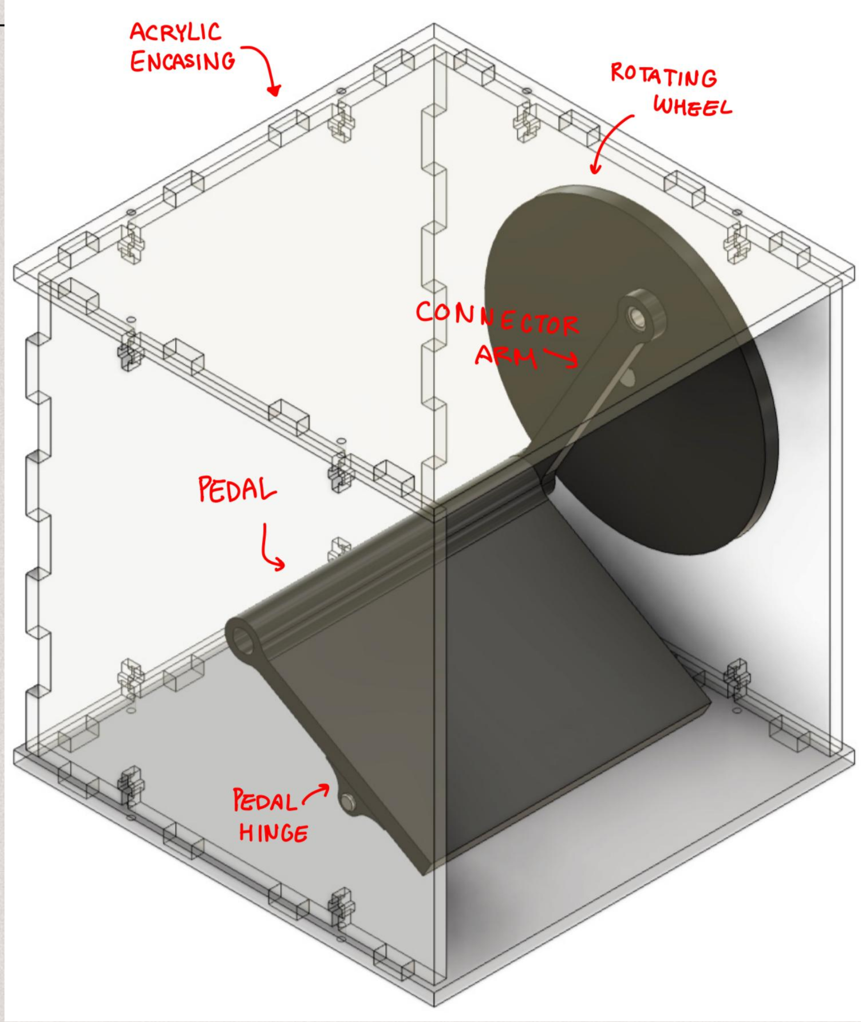


If pivot @ C & completely vertical

$\xrightarrow{+}$ $\Sigma F_x = F_{x,arm} \cdot l_{rod\ arm} + F_{x,wheel} \cdot r_{wheel\ radius} + (-F_{x,base}) \cdot l_{foot\ pedal} = 0$

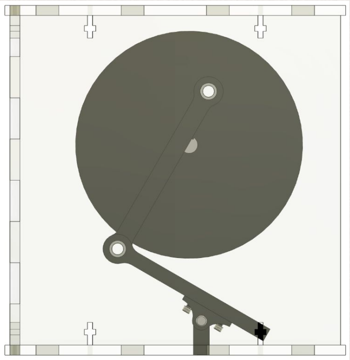
$\uparrow^{+}$ $\Sigma F_y = (-F_{y,base}) \cdot l_{foot\ pedal} + F_N \cdot l_{foot\ pedal} = 0$

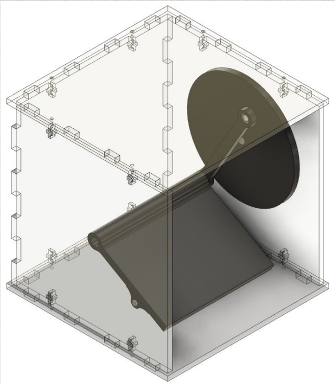
$\circlearrowright^{+}$ $\Sigma M = (-M_{base}) l_{foot\ pedal} + M_{wheel} \cdot r_{wheel\ radius} + M_{arm} \cdot l_{rod\ arm}$

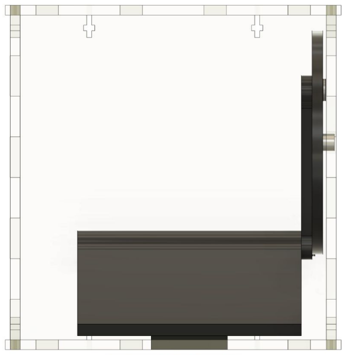


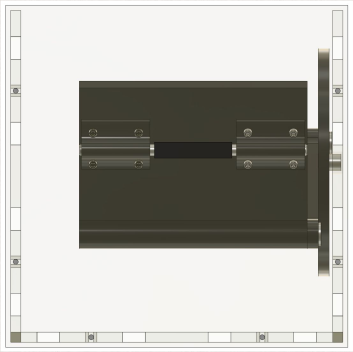
Building & Challenges

Building Process







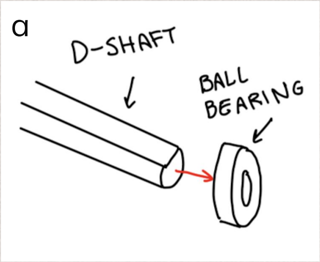


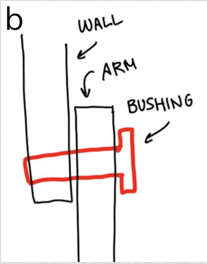
Notable features: Pedal hinge and dimensions, placements

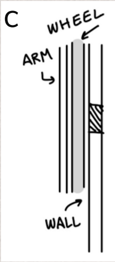
Main challenge: JOINTS
How to use parts to create the desired motion?
a. D-shaft was too tight to spin
b. Bushing was too tight, creating immobile friction
c. Determining how and where fasteners should go

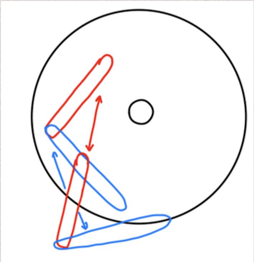
Secondary challenge: OVERALL PLACEMENT/LENGTHS

How long does the arm have to be, where should the wheel be, how steep should the angle on the foot pedal be?

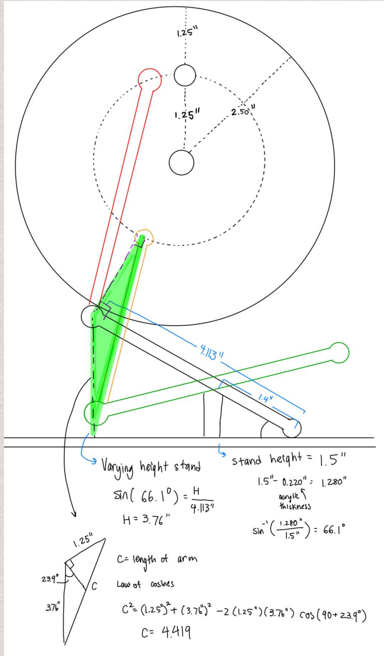








Right: An example set of calculations done for specific base height & arm length



Reflection

Takeaways

- The importance of having a **thorough design**, both on CAD and hand-drawn
  - Having started the design process early, throughout the week, we were able to add additional details that we had previously forgotten, such as screw holes, bushing dimensions, etc. It was significantly easier to communicate ideas when the specific part that we were working on was thoroughly hand drawn.
- A deep **understanding of joints & connections** helps smooth the process
  - We were both new to understanding how bushing and bearings connected, which made the assembly portion of the design process difficult. We realized the importance of having joints be an integrated part of the design process. It was significantly more difficult trying to figure out joints, fasteners, and other connections *after* things were designed
- **Assembly was difficult!**
  - Unlike Project 1, our mechanism required multiple moving parts to move together simultaneously. It was much more difficult to change one aspect of the mechanism because that would consequently change the entire assembly. It taught us to be intentional and careful with our changes.
- **Machines are inconsistent.**
  - We all learned this from Project 1, but it's nearly impossible to predict the kerfing in acrylic and heat expansion in 3D printing. Every print and cut resulted in varying levels of tolerances.

Future Improvements

- Make some kind of model sewing machine on the top of the acrylic covering/desk that we made to be even more use-case oriented
- Improve control on the direction of the spinning wheel
  - Make it so the placement of the pedal is able to go to a precise angle of the wheel. This would also allow for controlled forward/backward movement of the arms